Global Geometric-Entropic-Electromagnetic Model (GEE)

Unified Universe Operation Framework (No Dark Matter Alternative)

[Abstract]

This report presents a clean, flat-universe unified mathematical model designed to replace the speculative parameters of Dark Matter and Dark Energy in the standard Lambda-CDM model. Based on the foundational hypothesis that 'space exists only where energy waves and particles interact,' this framework integrates General Relativity, Statistical Thermodynamics (Information Entropy), and Relativistic Magnetohydrodynamics. The model resolves four major astronomical paradoxes—galaxy flattening, dark galaxies, bullet clusters, and cosmic microwave background asymmetry—without invoking unobservable shadow particles.

1. The GEE Unified Field Equation

The spatial geometry and cosmic structure of any region in the universe are governed self-consistently by the following four-term dynamic equilibrium equation:

$$G_{global} = R_{relativity} \pm \Phi_{aberration} + (k_B T / m) \cdot \nabla(\ln \Omega) + M_{(\theta,\phi)}$$

• Term 1: $R_{relativity}$ (Relativistic Mass-Energy Inertia)
The classic geometric term from General Relativity. It dominates high-acceleration, high-density local environments (e.g., Solar System, black hole accretion disks) where traditional mass matches observable gravity perfectly.

• Term 2: $\Phi_{aberration}$ (Gravitational Seesaw Aberration)
A geometric correction term that operates in inverse proportion to local acceleration. At deep space boundaries where gravity falls below the critical threshold ($\leq 1.2 \times 10^{-10}$ m/s^2), this term awakens to provide an extra geometric tension—acting as a 'gravitational balloon skin' that holds gas together without particle halos.

• Term 3: $(k_B T / m) \cdot \nabla(\ln \Omega)$ (Thermodynamic Information Entropy)
Where k_B is the Boltzmann constant, T is absolute temperature, and Ω represents the number of microscopic states. This term defines the active fabric of space and time. Matter naturally drifts toward maximum information entropy. In the high-temperature early universe, this gradient drives rapid structural seeding, while providing the expansion pressure that supports space from collapsing.

• **Term 4: $M_{(\theta,\phi)}$ (Anisotropic Electromagnetic Tensor)**
Governs the massive tension of primordial magnetic fields over millions of light-years. It incorporates directional beaming parameters (θ, ϕ), indicating that many cosmic entities emit energy like directed lasers. The fact that Earth-based telescopes miss the signals due to geometric misalignment does not mean matter is absent.

2. Paradigm Resolution & Validation

A. Galactic Thin-Disk Geometry: In the vertical axis (z-axis), matter collapses and particles collide, dissipating thermal energy via radiation. This process destroys the entropic gradient along the z-axis, causing the Φ term to shrink inward and flatten the galaxy. In the radial plane (r-axis), centrifugal inertia balances the standard R term, while the awakened Φ term locks the outer edges from dispersing. Finally, the toroidal magnetic lines of the M term align outlying anomalies into a precise plane.

B. Dark Galaxy Equilibrium (J1351+0039): Neutral hydrogen clouds move randomly at 11.6 km/s but maintain a perfect macroscopic sphere. GEE explains this as a perfect fluid-static balance between the outward kinetic expansion pressure and the inward tension of the awakened Φ term at ultra-low acceleration boundaries.

C. Bullet Cluster (1E 0657-56) Gravity Dislocation: During high-speed collision, dense ionized gas is halted in the center due to magnetohydrodynamic friction (M term under R rule). Conversely, stars pass through cleanly into sparse vacuums, where the lack of local density triggers an exponential surge in the inverse geometric Φ term. This surge bends space locally, causing the observed gravitational lensing centers to shift outward without needing ghost particles.

D. Cosmic Microwave Background (CMB) Peak Asymmetry: In the ultra-high temperature (high T) early universe, the entropic gradient term becomes immensely powerful. It amplifies the inward force during acoustic compression phases (odd peaks) and acts as a dampener during outward expansion phases (even peaks), perfectly replicating the observed acoustic power spectrum asymmetry.

3. The Boundary of Space: Energy Boundary Hypothesis

Space is not a pre-existing stage. Where energy waves (M), particles, and temperature ($T > 0$) reach, the physical equations activate to weave the fabric of 3D space. As the cosmic expansion front pushes into absolute nothingness, temperature drops toward 0 Kelvin. At $T=0$, the entropic term vanishes, the Φ term goes to geometric infinity, and incoming light beams freeze, losing all kinetic wave frequency. From an external perspective, the universe manifests as a sharp, non-reflective, frozen boundary layer where incoming light waves stack eternally as structural components of space itself.